

Henry Bao

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EDUCATION

University of Waterloo

Bachelor of Computer Science

Waterloo, ON

Sept. 2025 – Apr. 2030

TECHNICAL SKILLS

Languages: C/C++, Java, Python, JavaScript/Typescript, HTML/CSS, SQL, Golang, Rust, OCaml, Bash

Frameworks: Next.js, React, Node.js, Tailwind, Django, Flask, MediaPipe, Solana, FastAPI, RESTful APIs

Developer Tools: Git, VS Code, IntelliJ IDEA, CMake, Linux, Docker, Claude Code, Cursor, Terraform, Azure, Excel

Libraries: TensorFlow, Keras, PyTorch, scikit-learn, JAX, pandas, NumPy, Matplotlib, ffmpeg, PostgreSQL, Pytest

EXPERIENCE

Undergraduate Research Assistant

University of Waterloo

Sep. 2026 – Now

Waterloo, ON

- Working on practical concurrent ordered sets under Tom Iagovet

Infrastructure Support Specialist

Teranet

May 2026 – Aug. 2026

Toronto, ON

- Acted as the primary responder for alerts and incidents from Spectrum, Solarwinds, CommVault, etc.
- Fulfilled change requests for patches, server builds, and decommissions across various configurations
- Streamlined deployments and automated tasks using the infrastructure as code (IaC) tools Terraform and Ansible

PROJECTS

ExpVoxel | C++, CMake

Jul. 2026

- Developed a voxel ray tracer using hierarchical DDA traversal and AVX2 SIMD with Structure-of-Arrays layouts
- Built a lock-free Chase–Lev work-stealing scheduler using C++20 atomics for tile-based multithreaded rendering
- Used thread-local arena allocators and epoch-based reclamation, enabling low-overhead, concurrent world updates

ConstexprJIT | C++, CMake

Jul. 2026

- Built a C++23 compile-time JIT that emits executable x86-64 machine code from expressions using consteval
- Designed a constexpr pipeline and SSE2 backend to generate machine code without dynamic memory
- Wrote a runtime loader to manage relocation patching and dynamic symbol resolution for external function calls
- Implemented a memory manager with W^X-compliant protection to safely allocate and execute compiled code

Truv Compression | Rust

Jun. 2026

- Developed a compression tool based on the DEFLATE standard, utilizing LZ77 and dynamic Huffman coding
- Explored multi-threaded processing with rayon, partitioning files into independent blocks for faster compression
- Accelerated match finding by replacing byte-by-byte comparisons with 64-bit word operations through SWAR

Tiny Compiler | OCaml, Dune

Jun. 2026

- Built a functional language compiler supporting lexical scoping, first-class functions, conditionals, and let-bindings
- Implemented Hindley–Milner type inference (Algorithm W) with unification, polymorphic let-generalization, instantiation, and occurs-check detection for fully inferred static typing
- Wrote optimization passes for beta reduction, closure conversion, constant folding, and dead branch removal
- Created a stack-based virtual machine and bytecode engine to manage runtime execution and environment capture

p2p-file-sync | Golang, Claude Code

Feb. 2026

- Developed a decentralized file sync engine using Merkle Trees to reconcile state across peers in $O(\log n)$ time
- Reduced bandwidth and memory usage by implementing content-addressable storage with SHA-256 chunking
- Implemented a delta sync protocol using rolling hashes to isolate and transfer only the modified regions of a file
- Encrypted peer-to-peer networking over QUIC, using STUN-based UDP hole punching for NAT traversal

ACHIEVEMENTS

CEMC Contests: 14 honour rolls in mathematics and computing contests including one first-place and one runner-up

MAA Contests: 3 time qualification to the AIME mathematics contest

Advanced Coursework: 2x 4.0 and 4x 3.9 in advanced math and computer science courses

Certifications: Microsoft Certified: Azure Fundamentals (AZ-900) and HashiCorp Certified: Terraform Associate (004)